

Pedal harp improv guide

28 scales, 4 families, 7 modes each

How to use this guide

This guide walks through Oliver Prehn's 28-scale system, organized by his 4 families. Within each family, the 7 modes are presented in order from brightest (Lydian) to darkest (Locrian).

For each scale you'll find:

- what it sounds like and where you might hear it
- its characteristic tone (the note that defines its mood)
- the 7th chord that pairs with it
- a tip for improvising
- the pedal setting on tonic C, written in 4-cell braille

To play a scale on a different tonic, look up the row + column in the companion pedal matrix.

All 28 scales are equal in this system. You don't need to memorize any particular scale as more important than another. Browse by family, find scales whose sound you like, and add them to your vocabulary as you encounter them in pieces or improvisations.

Pedal braille notation

Each pedal setting is shown as 4 braille cells:

- cell 1: D pedal (left dots) + C pedal (right dots)
- cell 2: B pedal (left dots) + separator (always shows three vertical dots)
- cell 3: E pedal (left dots) + F pedal (right dots)
- cell 4: G pedal (left dots) + A pedal (right dots)

Within each cell, the dot row tells you the pedal position:

- top = flat
- middle = natural
- bottom = sharp

Example: ⠠⠠⠠⠠ = all naturals (C major scale).

Family 1: the major modes

Parent scale

Premise: Ionian
Popular: major scale
Pattern: W W H W W W H
(W=whole step, H=half step, W+H=whole-and-a-half step)

About this family

The 7 unaltered modes. The "white-key" scales. Every other family is built by altering one note in one of these modes.

Most tonal music lives here. If you don't know which family to pick, start in F1.

The 7 modes (brightest to darkest)

mode	pattern	character
Lydian	W W W H W W H	floating, dreamy, otherworldly
Ionian	W W H W W W H	happy, resolved, classic Western major
Mixolydian	W W H W W H W	rocky, swaggering, bluesy
Dorian	W H W W W H W	minor but optimistic; jazz, soul, Celtic music
Aeolian	W H W W H W W	sad, brooding, the default minor
Phrygian	H W W W H W W	Spanish, exotic, dark minor
Locrian	H W W H W W W	unstable, tense, queasy

Lydian

sound: floating, dreamy, otherworldly
use: film scores, cartoons, Joe Satriani; wonder/magic moments
mark tone: sharp 4 (raised tritone above tonic)
improv: *land on #4 as upper neighbor to 5; don't resolve #4→5*
chord: maj7 (C E G B on tonic C) but often play maj7#11
pattern: W W W H W W H
on C: C D E F# G A B
C pedals: ..♯..

Ionian

sound: happy, resolved, classic Western major
use: pop, folk, classical, hymns; anything that wants to feel "settled"
mark tone: natural 7 (leading tone resolves up to tonic)
improv: *use the leading tone 7→1 cadence; avoid lingering on 4*
chord: maj7 (C E G B on tonic C)
pattern: W W H W W W H
on C: C D E F G A B
C pedals: ..♯...

Mixolydian

sound: rocky, swaggering, bluesy
use: rock, blues, Celtic jigs, gospel
mark tone: flat 7 (no leading tone)
improv: *feature b7; cadence bVII→I instead of V→I*
chord: 7 (C E G Bb on tonic C, dominant 7)
pattern: W W H W W H W
on C: C D E F G A Bb
C pedals: ..:...

Dorian

sound: minor but optimistic; jazz, soul, Celtic music
use: "So What" by Miles Davis; modal jazz; medieval-feeling minor
mark tone: natural 6 in a minor mode (the brightener)
improv: *feature the natural 6 (A in C Dorian); IV chord is major*
chord: m7 (C Eb G Bb on tonic C)
pattern: W H W W W H W
on C: C D Eb F G A Bb
C pedals: ..:...

Aeolian

sound: sad, brooding, the default minor
use: pop ballads, folk laments, classical minor pieces
mark tone: flat 6 and flat 7 (both flat in a minor mode)
improv: *stay diatonic; raising 7 drifts into harmonic minor (F3)*
chord: m7 (C Eb G Bb on tonic C)
pattern: W H W W H W W
on C: C D Eb F G Ab Bb
C pedals: ..:...

Phrygian

sound: Spanish, exotic, dark minor
use: flamenco, metal, anything with Iberian or Middle-Eastern flavor
mark tone: flat 2 (semitone above tonic)
improv: *b2 is the signature; cadence bII→I; avoid V as arrival*
chord: m7 (C Eb G Bb on tonic C)
pattern: H W W W H W W
on C: C Db Eb F G Ab Bb
C pedals: ..:...

Locrian

sound: unstable, tense, queasy
use: rare as a tonic; appears in metal (Slayer), some Bartok
mark tone: flat 2 AND flat 5 (the only mode with diminished 5)
improv: *b5 makes the tonic chord diminished; use as color, not as home*
chord: m7b5 / ø (half-diminished)
pattern: H W W H W W W
on C: C Db Eb F Gb Ab Bb
C pedals: 

Family 2: the melodic minor modes

Parent scale

Premise: Dorian #7
Popular: ascending melodic minor
Pattern: W H W W W W H
(W=whole step, H=half step, W+H=whole-and-a-half step)

About this family

F1 with one sharp added. The alteration is different in each mode (Ionian #1, Dorian #7, etc.) but they all share the same 7-note set.

Jazz harmony lives here. Modern, sophisticated, "outside" colors over standard chord types.

The 7 modes (brightest to darkest)

mode	pattern	character
Lydian #5	W W W W H W H	floating Lydian plus augmented-chord tension
Ionian #1	H W H W W W W	maximum tension; alien, jazz "outside"
Mixolydian #4	W W W H W H W	dominant 7 chord with bright Lydian color
Dorian #7	W H W W W W H	minor with classical pull-to-tonic
Aeolian #3	W W H W H W W	major sound with sad b6 hanging over it
Phrygian #6	H W W W W H W	minor with bright 6 plus dark b2 – both characters at
Locrian #2	W H W H W W W	softer than plain Locrian; the standard "half-dim" s

Lydian #5 *(rotation of F2 parent at the Lydian degree)*

sound: floating Lydian plus augmented-chord tension
use: maj7#5 chords; whole-tone-adjacent moments in jazz
mark tone: Lydian with raised 5 (augmented chord at the root)
improv: *land on #4 and #5; both push outward from the major triad*
chord: maj7#5
pattern: W W W W H W H
on C: C D E F# G# A B
C pedals: 

Ionian #1 *(rotation of F2 parent at the Ionian degree)*

sound: maximum tension; alien, jazz "outside"
use: over altered dominants in jazz (V7alt resolving to I)
mark tone: every chord tone altered (b2 b3 b4 b5 b6 b7)
improv: every note clashes; lean into the dissonance
chord: 7alt (dominant with all alterations)
pattern: H W H W W W W
on C: C Db Eb Fb Gb Ab Bb
C pedals: ..:...

Mixolydian #4 *(rotation of F2 parent at the Mixolydian degree)*

sound: dominant 7 chord with bright Lydian color
use: common over 7#11 chords in jazz; "Simpsons theme"
mark tone: Mixolydian (b7) but with #4 as well
improv: feature b7 AND #4 – the trademark "Lydian dominant" pair
chord: 7#11
pattern: W W W H W H W
on C: C D E F# G A Bb
C pedals: ..:...

Dorian #7 *(rotation of F2 parent at the Dorian degree)*

sound: minor with classical pull-to-tonic
use: classical minor pieces ascending; jazz minor harmony
mark tone: minor scale with raised 7 (leading tone) AND raised 6
improv: the raised 7 wants to resolve up to tonic; combine with natural 6
chord: mMaj7 (minor with major 7th)
pattern: W H W W W W H
on C: C D Eb F G A B
C pedals: ..:...

Aeolian #3 *(rotation of F2 parent at the Aeolian degree)*

sound: major sound with sad b6 hanging over it
use: rare; sometimes over altered dominants
mark tone: natural minor with raised 3 – turns minor into a major-with-flat-6
improv: feels like a clouded major; lean on b6
chord: 7b13 (dominant with flat 13)
pattern: W W H W H W W
on C: C D E F G Ab Bb
C pedals: ..:...

Phrygian #6 *(rotation of F2 parent at the Phrygian degree)*

sound: minor with bright 6 plus dark b2 – both characters at once
use: jazz over minor chords needing extra color
mark tone: Dorian-bright minor with a flat 2
improv: *feature both natural 6 (Dorian color) and b2 (Phrygian color)*
chord: m7 (with b9 tension available)
pattern: H W W W W H W
on C: C Db Eb F G A Bb
C pedals: 

Locrian #2 *(rotation of F2 parent at the Locrian degree)*

sound: softer than plain Locrian; the standard "half-dim" sound
use: over m7b5 chords in jazz minor 2-5-1
mark tone: Locrian but with natural 2 (less alien than plain Locrian)
improv: *works wonderfully over half-diminished chords; b5 is the marker*
chord: m7b5 / ø (half-diminished 7)
pattern: W H W H W W W
on C: C D Eb F Gb Ab Bb
C pedals: 

Family 3: the harmonic minor modes

Parent scale

Premise: Aeolian #7
Popular: harmonic minor
Pattern: W H W W H W+H H
(W=whole step, H=half step, W+H=whole-and-a-half step)

About this family

F1 with a different sharp. Contains a wide whole-and-a-half step that gives this family its distinctive "exotic" sound.

Classical minor cadences (Bach), flamenco, klezmer, Middle-Eastern flavors, metal.

The 7 modes (brightest to darkest)

mode	pattern	character
Lydian #2	W+H H W H W W H	Lydian dreaminess plus a sharp opening interval
Ionian #5	W W H W+H H W H	major with augmented-chord tension
Mixolydian #1	H W H W W H W+H	unusual; rarely used except as theoretical curiosity
Dorian #4	W H W+H H W H W	minor with an exotic raised 4; Eastern European flavor
Aeolian #7	W H W W H W+H H	classical minor; dramatic, "European" minor
Phrygian #3	H W+H H W H W W	flamenco, Andalusian, Middle-Eastern
Locrian #6	H W W H W+H H W	half-diminished feel with extra brightness on 6

Lydian #2 *(rotation of F3 parent at the Lydian degree)*

sound: Lydian dreaminess plus a sharp opening interval
use: jazz over maj7#9 or maj7#11 chords; rare elsewhere
mark tone: Lydian with raised 2 (gives a wide step from root to #2)
improv: *the wide root-to-#2 step gives this its character*
chord: maj7 (with #9 color)
pattern: W+H H W H W W H
on C: C D# E F# G A B
C pedals: .:~:..

Ionian #5 *(rotation of F3 parent at the Ionian degree)*

sound: major with augmented-chord tension
use: over maj7#5 chords; rare in tonal music
mark tone: major scale with raised 5 (augmented chord at the root)
improv: *the raised 5 sits between 5 and 6 of the major scale*
chord: maj7#5
pattern: W W H W+H H W H
on C: C D E F G# A B
C pedals: ..:~:.

Mixolydian #1 *(rotation of F3 parent at the Mixolydian degree)*

sound: unusual; rarely used except as theoretical curiosity
use: mostly appears as the 5th mode of harmonic minor
mark tone: Mixolydian with raised root – creates exotic intervals
improv: *rarely used as a standalone scale*
chord: °7 (diminished 7, since the root is altered)
pattern: H W H W W H W+H
on C#: C# D E F G A Bb
C# pedals: .:~:...

Dorian #4 *(rotation of F3 parent at the Dorian degree)*

sound: minor with an exotic raised 4; Eastern European flavor
use: klezmer, Romani music, some film scores
mark tone: Dorian with raised 4 (the wide step from b3 to #4)
improv: *feature the wide step b3 → #4 – the signature interval*
chord: m7 (with #11 color available)
pattern: W H W+H H W H W
on C: C D Eb F# G A Bb
C pedals: ..:~:..

Aeolian #7 *(rotation of F3 parent at the Aeolian degree)*

sound: classical minor; dramatic, "European" minor
use: Bach, Mozart, classical minor cadences; metal; flamenco modulations
mark tone: natural minor with raised 7 (leading tone)
improv: *the raised 7 creates the V→i cadence missing in plain Aeolian*
chord: mMaj7 (minor with major 7th)
pattern: W H W W H W+H H
on C: C D Eb F G Ab B
C pedals: 

Phrygian #3 *(rotation of F3 parent at the Phrygian degree)*

sound: flamenco, Andalusian, Middle-Eastern
use: Spanish music, klezmer, metal (Yngwie Malmsteen); over V7b9 in minor keys
mark tone: Phrygian with raised 3 – produces dominant 7 chord at the root
improv: *the wide step from b2 to natural 3 is the signature; lean on b2 and natural 3*
chord: 7b9
pattern: H W+H H W H W W
on C: C Db E F G Ab Bb
C pedals: 

Locrian #6 *(rotation of F3 parent at the Locrian degree)*

sound: half-diminished feel with extra brightness on 6
use: jazz over m7b5 chords as alternative to Locrian b2
mark tone: Locrian but with natural 6 (raised from plain Locrian's b6)
improv: *similar to Locrian b2 in use; pick by which extension you want*
chord: m7b5 (half-diminished)
pattern: H W W H W+H H W
on C: C Db Eb F Gb A Bb
C pedals: 

Family 4: the harmonic major modes

Parent scale

Premise: Ionian b6
Popular: harmonic major
Pattern: W W H W H W+H H
(W=whole step, H=half step, W+H=whole-and-a-half step)

About this family

F1 with one flat added. Like F3 but built from a major (Ionian) base instead of minor (Aeolian). Contains the same wide whole-and-a-half step.

Mysterious major-key colors; film scores; Hollywood "Eastern" pastiche; some 20th-century classical.

The 7 modes (brightest to darkest)

mode	pattern	character
Lydian b3	W H W+H H W W H	minor mode with raised 4 – exotic minor
Ionian b6	W W H W H W+H H	major with a darkened 6th; "wistful major"
Mixolydian b2	H W+H H W W H W	dominant 7 chord with Phrygian-flavored upper structure
Dorian b5	W H W H W+H H W	half-diminished feel with Dorian's natural 6
Aeolian b1	W+H H W W H W H	used as a rotation rather than as a tonic scale
Phrygian b4	H W H W+H H W W	darker Phrygian; ultra-exotic
Locrian b7	H W W H W H W+H	fully diminished, very dark

Lydian b3 *(rotation of F4 parent at the Lydian degree)*

sound: minor mode with raised 4 – exotic minor
use: jazz fusion; some film score textures
mark tone: Lydian with flat 3 – turns the major into minor with #4
improv: #4 in a minor mode creates a haunting quality
chord: m(maj7) with #11
pattern: W H W+H H W W H
on C: C D E $\flat$ F $\sharp$ G A B
C pedals:

Ionian b6 *(rotation of F4 parent at the Ionian degree)*

sound: major with a darkened 6th; "wistful major"
use: film scores; some classical (Brahms); modal jazz
mark tone: major scale with flat 6 (borrowed from minor)
improv: the b6 is the modal marker; use it to add wistfulness to major
chord: maj7 (with b13 color available)
pattern: W W H W H W+H H
on C: C D E F G A $\flat$ B
C pedals:

Mixolydian b2 *(rotation of F4 parent at the Mixolydian degree)*

sound: dominant 7 chord with Phrygian-flavored upper structure
use: jazz over 7b9 chords resolving to I
mark tone: Mixolydian (b7) with flat 2
improv: *similar to Phrygian dominant but with natural 6*
chord: 7b9
pattern: H W+H H W W H W
on C: C Db E F G A Bb
C pedals: 

Dorian b5 *(rotation of F4 parent at the Dorian degree)*

sound: half-diminished feel with Dorian's natural 6
use: jazz over m7b5 with bright extensions
mark tone: Dorian with flat 5 – turns the minor into a half-diminished color
improv: *natural 6 brightens this half-dim variant*
chord: m7b5
pattern: W H W H W+H H W
on C: C D Eb F Gb A Bb
C pedals: 

Aeolian b1 *(rotation of F4 parent at the Aeolian degree)*

sound: used as a rotation rather than as a tonic scale
use: mostly theoretical; appears as 6th mode of harmonic major
mark tone: Aeolian with flattened root creates unusual intervals
improv: *rarely used as standalone; usually a passing rotation*
chord: (no clean chord-scale match)
pattern: W+H H W W H W H
on C: C D# E F# G# A B
C pedals: 

Phrygian b4 *(rotation of F4 parent at the Phrygian degree)*

sound: darker Phrygian; ultra-exotic
use: rare; appears in flamenco and some Bartok
mark tone: Phrygian with flat 4 – creates wide step from 4 to 5
improv: *the b2 + b4 combination is unusual; treat as color*
chord: m7 (with b11 dissonance)
pattern: H W H W+H H W W
on C: C Db Eb Fb G Ab Bb
C pedals: 

Locrian b7 *(rotation of F4 parent at the Locrian degree)*

sound: fully diminished, very dark

use: over diminished 7 chords in jazz and classical

mark tone: Locrian with further-flattened 7 – creates diminished 7 chord

improv: *the bb7 is enharmonically a 6 – but in this naming, the chord built up is dim7*

chord: °7 (diminished 7)

pattern: H W W H W H W+H

on C#: C# D E F# G A Bb

C# pedals: · · · · ·

Appendix: brightness ordering

Within each family, modes go brightest (Lydian) to darkest (Locrian).
Across families, brightness depends on which alterations cancel or compound.

Sequence within each family:

Lydian → Ionian → Mixolydian → Dorian → Aeolian → Phrygian → Locrian

Family character (rough):

- F1 neutral; the unaltered diatonic baseline
- F2 brighter where alteration sharps a note (Lydian #5 is very bright);
 darker where alteration is in already-dark mode (Locrian #2)
- F3 exotic; the wide W+H step adds a 'foreign' color
- F4 darker than F1 because adding a flat darkens

Appendix: pedal matrix – F1/F2 x 12 tonics

Cells: D-C || B-sep || E-F || G-A Per cell: top=flat, middle=natural, bottom=sharp * = enharmonic respelling

	C	G	D	A	E	B	F#	F	Bb	Eb	Ab	Db	Gb
F1 major													
Ionian	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••
Dorian	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••*
Phrygian	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••*	•••••*	•••••*
Lydian	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••
Mixolydian	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••
Aeolian	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••*	•••••*
Locrian	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••*	•••••*	•••••*	•••••*
F2 melodic minor													
Ionian #1	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••*	•••••*	•••••*	•••••*	•••••*	•••••*
Dorian #7	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••*
Phrygian #6	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••*	•••••*	•••••*
Lydian #5	•••••	•••••	•••••	•••••	•••••	•••••*	•••••*	•••••	•••••	•••••	•••••	•••••	•••••
Mixolydian #4	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••
Aeolian #3	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••*	•••••*
Locrian #2	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••	•••••*	•••••*	•••••*	•••••*

Appendix: pedal matrix – F3/F4 x 12 tonics

Cells: D-C || B-sep || E-F || G-A Per cell: top=flat, middle=natural, bottom=sharp * = enharmonic respelling

	C	G	D	A	E	B	F#	F	Bb	Eb	Ab	Db	Gb
F3 harmonic minor													
Ionian #5	..♯..	.♯..	.♯..	.♯..	..♯..	..♯..	* ..♯..	* ..♯..	.♯..	..♯..	.♯..	.♯..	.♯..
Dorian #4	..♯..	.♯..	..♯..	.♯..	.♯..	.♯..	..♯..	..♯..	.♯..	.♯..	.♯..	..♯..	..♯..*
Phrygian #3	.♯..	..♯..	..♯..	.♯..	..♯..	.♯..	.♯..	.♯..	..♯..	..♯..	..♯..	..♯..*	.♯..*
Lydian #2	.♯..	.♯..	.♯..	..♯..	..♯..	* ..♯..	* ..♯..	* ..♯..	..♯..	.♯..	..♯..	.♯..	.♯..
Mixolydian #1	..♯..*	* ..♯..	* ..♯..	.♯..	.♯..	..♯..	..♯..	.♯..*	.♯..*	.♯..*	..♯..*	.♯..*	..♯..*
Aeolian #7	..♯..	..♯..	.♯..	..♯..	.♯..	.♯..	.♯..	.♯..	.♯..	.♯..	..♯..	..♯..*	.♯..*
Locrian #6	.♯..	.♯..	..♯..	..♯..	.♯..	..♯..	.♯..	..♯..	..♯..	..♯..*	.♯..*	.♯..*	.♯..*
F4 harmonic major													
Ionian b6	..♯..	..♯..	.♯..	.♯..	.♯..	..♯..	.♯..	.♯..	.♯..	.♯..	.♯..	..♯..*	.♯..*
Dorian b5	..♯..	.♯..	..♯..	..♯..	.♯..	.♯..	.♯..	.♯..	.♯..	..♯..*	.♯..*	..♯..*	.♯..*
Phrygian b4	.♯..	.♯..	..♯..	.♯..	..♯..	..♯..	.♯..	..♯..*	.♯..*	..♯..*	.♯..*	.♯..*	.♯..*
Lydian b3	..♯..	.♯..	.♯..	.♯..	..♯..	.♯..	..♯..	..♯..	.♯..	..♯..	.♯..	.♯..	..♯..*
Mixolydian b2	.♯..	..♯..	..♯..	.♯..	.♯..	.♯..	..♯..	..♯..	.♯..	.♯..	..♯..*	.♯..*	..♯..*
Aeolian b1	.♯..	..♯..	.♯..	..♯..	..♯..	* ..♯..	* ..♯..	* ..♯..	.♯..	.♯..	..♯..	.♯..	..♯..
Locrian b7	..♯..*	* ..♯..	* ..♯..	.♯..	.♯..	..♯..	..♯..	.♯..*	.♯..*	.♯..*	.♯..*	.♯..*	..♯..*